DATE: 4/29/2026 **AUTHOR**: Andy Creque **SUBJECT**: R-Zero Take-Home Assignment

Design Overview:

I created this document before diving into coding. The design pattern I chose separates data acquisition and transmission by putting a queue in between. The data acquisition logic drops new data in the queue continuously regardless of network state. When the network is up, data is transmitted as fast as it is acquired – only one slot in the queue is used. When the network goes down, the acquisition logic continues filling queue slots without interruption. Since network transmission is several orders of magnitude faster than data acquisition, so the network drains the queue quickly once it comes back up.

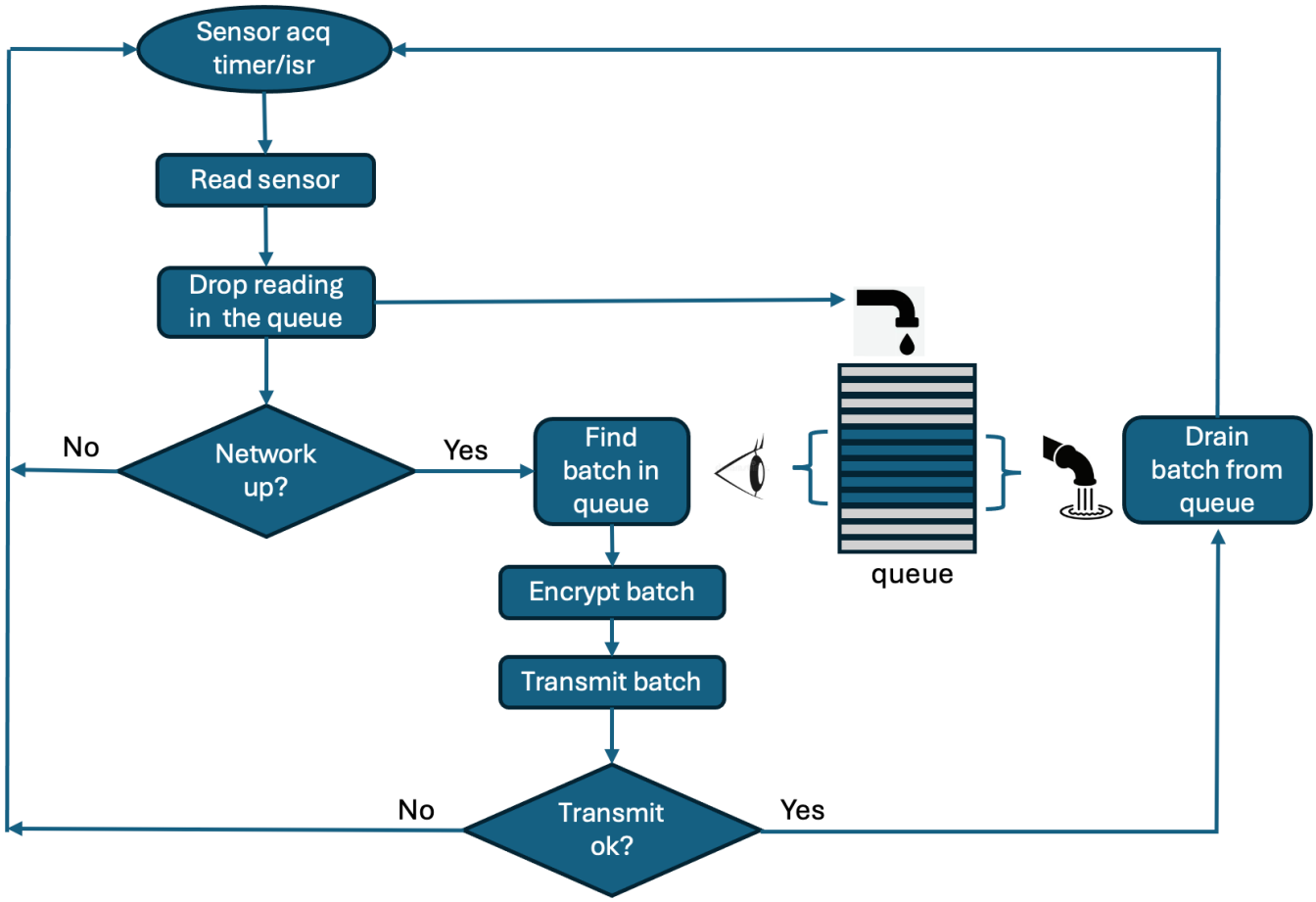
Core Assumptions/Tradeoffs:

1. The queue will be circular. When the queue becomes full and wraps around, the oldest data in the queue is overwritten and lost. I assume the most recent occupancy data is more important than old data.
2. A sensor reading will be 8 bytes and include occupancy count, timestamp, and a unique sequence number. If the queue is 1024 bytes, the amount of time the network can be down without data loss would be:

$$hours = \frac{1024\ bytes}{(8\frac{\text{bytes}}{\text{reading}} * 6\frac{readings}{min} * 60\frac{min}{hour}} = 0.35\ hours \rightarrow 21\ minutes$$

3. Queued data is not stored in non-volatile memory so if power is lost, all data in the queue is lost.
4. Data will be encrypted to prevent snooping but done so using an algorithm that also supports data authentication (i.e., detects when data has been altered or corrupted in flight).

NOTE: I'm good a PowerPoint so after a napkin sketch, I was able to create this in about 30 minutes. Kind of overkill, but I might leverage this in the future for real work 😉



a. New data always dropped into the queue and oldest data is overwritten if no empty slots.
b. Data is transmitted in batches comprised of one or more data slots.
c. Finding a batch in the queue, or pulling a batch from the queue for transmission, does not affect the queue head and tail pointers.
d. Batch data is drained from the queue AFTER successful transmission.
e. Should use a mutex to protect the queue head and tail pointers when pushing or erasing queue data.